The influence of functional differences on the evolution of speech and music transmission range

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Speech and singing are the two most complex forms of vocalization unique to *Homo sapiens*. Due to their shared features, some scholars have proposed a common origin for language and music (Bannan, 2008; Brown, 2000, 2017; Darwin, 1871; Fitch, 2005, 2010, 2013). Both rely primarily on the auditory domain for transmission and share affective prosody (Scherer, 2013)—perhaps the most evident similarity. However, despite these overlaps, language and music differ significantly in their functions and usage. Language is widely regarded as the primary tool for exchanging propositional meaning between individuals (Deacon, 1997; Jackendoff, 2002; Millikan, 2005) and, according to some scholars, primarily as the medium for thought (Chomsky, 2007). Music, by contrast, lacks these characteristics (Lerdahl, 2013). It is typically performed collectively and simultaneously, whereas speech usually occurs in dialogic form, with alternating sequences of sound. There are also notable structural differences. For instance, pitch in singing tends to be more stable than in speech (Raposo de Medeiros et al., 2021; Zatorre & Baum, 2012), a pattern observed across cultures (Ozaki et al., 2024). Musical rhythm is generally more regular (London, 2012) and music itself more repetitive (Margulis, 2014) than spoken language. These structural and functional distinctions are often interpreted as outcomes of different evolutionary pressures that shaped these two modes of human communication. Nonetheless, debate continues regarding the specific adaptive values that led to the emergence of music and language as distinct communicative tools. One influential hypothesis contrasts the choric nature of music with the dialogic nature of language (Haiduk & Fitch, 2022), suggesting that music evolved to foster social bonding (Dunbar, 2012; Pearce et al., 2015; Roederer, 1984; Savage et al., 2021), while language developed to facilitate the exchange of factual information and ideas (Deacon, 1997; Jackendoff, 2002; Millikan, 2005).

This proposal aims to highlight another often overlooked factor that may have played a crucial role in the divergent evolution of language and music in *Homo sapiens*: transmission range. This factor may have influenced primarily the vocal-motor domain rather than the evolution of language as a cognitive system. It is hypothesized that hominins, like other primates (Waser & Waser, 1977),

employed two functionally distinct types of vocal signaling—intergroup long-distance calls and intragroup short-distance calls—each shaped by the acoustic constraints of their environment. Signals such as territorial calls are only adaptive if audible over long distances, whereas others, like covert conversations, confer evolutionary advantages when they are less conspicuous. Many nonhuman primate species that engage in song-like vocalizations produce both long-distance and short-distance calls. Natural selection favored different acoustic features for the distinct signaling functions related to long-distance versus short-distance calls. This scenario is supported by several differences between language and music that relate to signal propagation. Features associated with the choric versus dialogic nature of music and language are directly linked to transmission range. For example, group singing or drumming amplifies the acoustic signal, increasing its audibility over distance. Collective musical performance often involves rhythmic synchronization (Patel, 2006) and pitch matching (Bannan, 2012, 2020), both of which additionally enhance long-range signal propagation (Hagen & Bryant, 2003; Hagen & Hammerstein, 2009; Podlipniak, 2025). A clear example of a vocal adaptation for long-distance calls is found in gibbon songs, in which the first formant is precisely tuned to the fundamental frequency (F0), thereby amplifying exclusively the F0 component of the source signal. This technique is very similar to the one used by professional human soprano singers (Koda et al., 2012).

Equally important as range is signal legibility. In music, information is primarily encoded through rhythm and pitch, and the stability of the fundamental frequency (F0) makes the signal less vulnerable to attenuation and reverberation. In contrast, spoken language relies mainly on amplitude modulation (Shannon et al., 1995), which limits its clarity over long distances. Particularly compelling in this context are examples of paralinguistic systems designed for long-distance communication, such as whistled (Meyer, 2008, 2015) and (Seifart et al., 2018), and shepherd calls like Kulning (Rosenberg et al., 2024). In these cases, musical elements are used to encode linguistic content. Moreover, music's greater repetitiveness compared to speech introduces redundancy (Margulis, 2014), which aids in signal recognition when attenuation and reverberation occur—common challenges in long-distance transmission due to wind and environmental obstacles.

Taken together, these differences suggest that the adaptive function of music is closely tied to intergroup long-distance communication, whereas language primarily serves intragroup short-distance interaction.

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